

SOC Internship – Day 28 Notes

VirusTotal

What is VirusTotal?

VirusTotal is a free online threat intelligence platform that analyzes files, URLs, domains, and IP addresses using multiple antivirus engines and security tools. It helps security professionals identify malware, phishing links, and suspicious activities.

VirusTotal

Learning Objectives

- Understand the purpose of VirusTotal.
- Learn how file and URL scanning works.
- Identify suspicious indicators in security reports.
- Develop basic cybersecurity analysis skills.

Key Features

1. File Scanning

- Upload files for malware analysis.
- Scanned by multiple antivirus engines.
- Provides detection results and threat information.

2. URL Analysis

- Checks websites for malicious content.
- Detects phishing, malware distribution, and suspicious behavior.

3. Domain & IP Reputation

- Analyzes domain names and IP addresses.
- Shows historical and reputation-related information.

4. Community Intelligence

- Security researchers share comments and findings.
- Helps validate threats using community knowledge.

Important Indicators in a VirusTotal Report

Detection Ratio

- Shows how many security engines flagged the file.
- Example: **5/70** means 5 engines detected the file as malicious.

File Hash

- Unique identifier of a file.
- Common hash types:
 - MD5
 - SHA-1
 - SHA-256

Scan Engine Results

- Displays verdicts from different antivirus vendors.
- Helps compare detection accuracy across engines.

Internship Tasks Performed

1. Explored the VirusTotal dashboard.
2. Uploaded a sample file and reviewed the analysis.
3. Analyzed a URL for security risks.
4. Examined:
 - a. Detection Ratio
 - b. File Hash
 - c. Scan Engine Results
5. Documented findings in a summary report.
6. Collected screenshots for submission.

Sample Summary Report (100–150 Words)

VirusTotal was used to analyze a sample file and a URL to understand how threat detection works. The platform scanned the file using multiple antivirus engines and provided detailed results, including the detection ratio, file hash values, and individual scan engine verdicts. The URL analysis helped identify whether the website contained malicious or suspicious content. The detection ratio provided a quick overview of the file's reputation, while the hash values served as unique identifiers for verification and tracking. The scan engine results demonstrated how different security vendors classify threats. Overall, VirusTotal proved to be a valuable cybersecurity tool for malware analysis, threat intelligence, and reputation checking, helping analysts make informed decisions about potentially harmful files and websites.

